(PV-TM-01) Traceability Matrix

Document ID: `PV-TM-01`
Product: `Portview`
Document Status: `Released`

Document Overview

This document records the traceability relationships used across the Portview software document set.

Document notes:

- The current matrix content shows requirement and risk linkage and should be normalized into a stable authored format.
- ImageProcess-specific rows are excluded from the current authored scope.

Document Approval

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Revision History

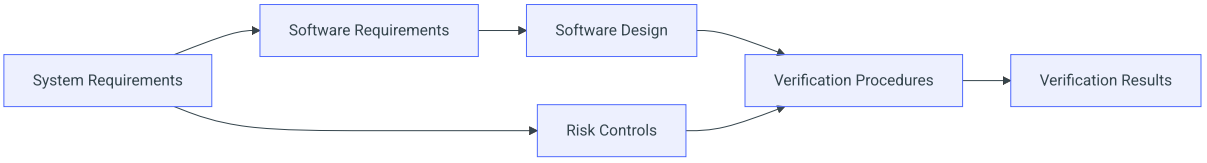
Rev.	Date	Description
`0.1`	`To confirm`	Initial authored traceability matrix entry pending normalization

1. Purpose

This document provides traceability across Portview requirements, risk controls, design allocation, and verification coverage.

2. Traceability Model

The intended traceability chain is:



3. Current Coverage Areas

The current matrix content shows linkage in the following areas.

Traceability Area	Relationship
Risk to requirement	Risk items are linked to system or software requirements
Requirement to verification	Requirements are linked to one or more verification activities
Risk to verification	Risk-control verification is visible through linked requirement coverage
Design to verification	Design allocation is expected to be linked to verification evidence in the authored set

4. Portview-Oriented Matrix Content

The current matrix content visibly covers:

- acquisition and sensor workflow
- patient management and search behavior
- annotation and measurement tools
- DICOM print and DICOM communication
- export and media handling
- device connection and integrity checks
- logging and instability guidance

4.1 Observed Risk-To-Requirement Themes

The currently visible matrix rows indicate recurring traceability themes.

Trace Theme	Typical Relationship
Acquisition failure	sensor or acquisition failure linked to acquisition and device requirements
Patient-data workflow failure	patient registration, search, and retrieval linked to account or patient-management requirements
Annotation and measurement failure	annotation hazards linked to viewer-tool requirements
Output and communication failure	print, DICOM, export, and media hazards linked to output requirements
Device integrity and connectivity failure	USB, driver, and network issues linked to integrity and connectivity requirements
Stability and data-handling failure	logging, instability guidance, and transmission problems linked to robustness requirements

4.2 Coverage Boundaries

The authored Portview matrix should keep the following scope boundaries.

- include Portview system and software requirements
- include Portview design allocation and verification references
- include risk items that map directly to Portview workflow or control behavior

- exclude ImageProcess-only processing controls unless explicitly added back into scope

5. Authored Matrix Structure

The normalized matrix should use at least the following columns.

Column	Purpose
Risk Item	Identifies the hazardous situation, failure mode, or control context
Requirement ID	Identifies the authored system or software requirement addressed
Design Allocation	Identifies the module or design item implementing the requirement
Procedure ID	Identifies the verification procedure used
Result Record	Identifies the verification evidence or result record
Status	Identifies whether coverage is complete, partial, or pending

5.1 Operating Environment And Access

RS ID	SRS ID	SDS ID	UTP ID	ITP ID	Risk Item	Status
`RS-001`	`SRS-001`	`SDS-001`	`UTP-024`	`ITP-002`	`-`	`Linked`
`RS-002`	`SRS-036`	—	—	`ITP-001`	`FR-004` `context`	`Linked`

5.2 Acquisition And Device Interaction

RS ID	SRS ID	SDS ID	UTP ID	ITP ID	Risk Item	Status
`RS-003`	`SRS-006`	`SDS-024`	`UTP-001`	`ITP-008`	`FR-018`	`Linked`
`RS-004`	`SRS-009`	`SDS-002`	`UTP-002`	`ITP-010`	`FR-014` `context`	`Linked`

RS ID	SRS ID	SDS ID	UTP ID	ITP ID	Risk Item	Status
RS-005	SRS-010	SDS-025	UTP-003	ITP-011	—	Linked
RS-006	SRS-030	SDS-027	UTP-018	ITP-032	—	Linked
RS-007	SRS-007, SRS-031	SDS-023, SDS-028	UTP-005, UTP-019	ITP-007, ITP-031	FR-012	Linked
RS-008	SRS-008	SDS-026	UTP-004	ITP-012	FR-001 context	Linked
RS-009	SRS-008, SRS-032	SDS-026, SDS-029	UTP-004, UTP-020	ITP-012	FR-012 context	Linked

5.3 Viewing And Diagnostic Interaction

RS ID	SRS ID	SDS ID	UTP ID	ITP ID	Risk Item	Status
RS-010	SRS-011	SDS-003	UTP-006	ITP-013	FR-002	Linked
RS-010	SRS-012	SDS-004	UTP-012	ITP-014	FR-002 context	Linked
RS-011	SRS-013	SDS-005	UTP-007	ITP-016	—	Linked
RS-012	SRS-014	SDS-006	UTP-008	ITP-015	-	Linked
RS-013	SRS-015	SDS-003	UTP-009	ITP-017	—	Linked
RS-013	SRS-016	SDS-013	UTP-010	ITP-018	—	Linked
RS-013	SRS-017	SDS-014	UTP-011	ITP-021	—	Linked
RS-014	SRS-018	SDS-007	—	ITP-022	FR-013	Integration only
RS-014	SRS-019	SDS-008	—	ITP-022	FR-013	Integration only
RS-014	SRS-020	SDS-009	—	ITP-023	FR-013	Integration only

RS ID	SRS ID	SDS ID	UTP ID	ITP ID	Risk Item	Status
RS-014	SRS-021	SDS-010	—	ITP-024	FR-013	Integration only
RS-014	SRS-022	SDS-011	—	ITP-025	FR-013	Integration only
RS-014	SRS-023	SDS-012	—	ITP-026	FR-013	Integration only

5.4 Output And Communication

RS ID	SRS ID	SDS ID	UTP ID	ITP ID	Risk Item	Status
RS-015	SRS-024	SDS-015	UTP-013	ITP-027	-	Linked
RS-015	SRS-025	SDS-016	UTP-014	ITP-028	FR-011	Linked
RS-016	SRS-026	SDS-017	UTP-015	ITP-006	FR-010	Linked
RS-017	SRS-027	SDS-018	UTP-016	ITP-029	FR-009	Linked
RS-018	SRS-028	SDS-019	UTP-017	ITP-030	FR-006, FR-008	Linked
—	SRS-029	SDS-002	UTP-002	ITP-010	—	Linked

5.5 Integrity, Robustness, And User Guidance

RS ID	SRS ID	SDS ID	UTP ID	ITP ID	Risk Item	Status
RS-019	SRS-033	SDS-030	UTP-021	ITP-033	FR-007, FR-012	Linked
RS-020	SRS-034	SDS-020	UTP-022	ITP-034	FR-008 context	Linked
RS-021	SRS-035	SDS-021	UTP-023	ITP-035	FR-004	Linked
RS-022	SRS-036	—	—	ITP-001	FR-004 context	Linked

RS ID	SRS ID	SDS ID	UTP ID	ITP ID	Risk Item	Status
RS-023	SRS-036	—	—	ITP-001	FR-001, FR-012 context	Linked

5.6 Administrative And Localization

RS ID	SRS ID	SDS ID	UTP ID	ITP ID	Risk Item	Status
RS-024	SRS-002	SDS-001	UTP-024	ITP-002	FR-017	Linked
RS-024	SRS-003	SDS-001	UTP-024	ITP-003	FR-017	Linked
RS-024	SRS-004	SDS-001	UTP-024	ITP-004	FR-015	Linked
RS-024	SRS-005	SDS-001	UTP-024	ITP-005	FR-016	Linked
RS-025	SRS-037	SDS-022	UTP-025	ITP-036	-	Linked

5.7 System-Level Release Verification

RS ID	System Procedure ID	System Result ID	Risk Item	Status
RS-001, RS-002	SYSP-001	PV-SYSTR-01	FR-001, FR-002 context	Linked at system level
RS-003 to RS-009, RS-024	SYSP-002	PV-SYSTR-01	FR-012, FR-017, FR-018	Linked at system level
RS-010 to RS-014	SYSP-003	PV-SYSTR-01	FR-001, FR-002, FR-013	Linked at system level
RS-015 to RS-018	SYSP-004	PV-SYSTR-01	FR-006, FR-008, FR-009, FR-010, FR-011	Linked at system level
RS-019, RS-020, RS-022, RS-023	SYSP-005	PV-SYSTR-01	FR-007, FR-012	Linked at system level

RS ID	System Procedure ID	System Result ID	Risk Item	Status
`RS-021`, `RS-025`	`SYSP-006`	`PV-SYSTR-01`	`FR-004 context`	`Linked at system level`
`RS-001`, `RS-019`, `RS-020`, `RS-023`	`SYSP-007`	`PV-SYSTR-01`	`FR-003`, `FR-005`, `FR-007`	`Linked at system level`
`RS-021`, `RS-023`	`SYSP-008`	`PV-SYSTR-01`	`FR-004`	`Linked at system level`
`RS-003`, `RS-010`, `RS-016`, `RS-024`	`SYSP-009`	`PV-SYSTR-01`	`FR-014`, `FR-015`, `FR-016`, `FR-017`, `FR-018 context`	`Linked at system level`

5.8 Coverage Notes

- Annotation and measurement requirements (`SRS-018` to `SRS-023`) are verified at integration level only (`ITP-022` to `ITP-026`); no separate unit procedures exist for these items. **Design decision justification**: Annotation and measurement tools do not operate independently. They require an active patient context, a loaded image set, and a rendered mount state to function. Verifying these tools in isolation at unit level would require simulating the entire viewer environment, producing less meaningful results than verifying them within the integrated viewer workflow where they actually operate. This verification-level allocation is a deliberate design decision under IEC 62304 Class B, where the standard permits justified selection of the appropriate verification level.
- Device Service unit functions are now allocated to individually numbered design items (`SDS-023` to `SDS-030`) in `PV-SDS-01` . The previous design-allocation gap has been resolved.
- System-level verification should be linked through `PV-SYSTP-01` SystemTP-Z01 and `PV-SYSTR-01` SystemTR-Z01.
- Unit-result linkage should use `PV-STR-01` SwSTR-Z01 for Portview.
- Integration-result linkage should use `PV-TR-01` SwTR-Z01 for Portview.

6. Coverage By Document Set

The authored matrix should connect the current Portview document set as follows.

Source Document	Matrix Role
`PV-RS-01` RS for Portview	System-level source requirements
`PV-SRS-01` SwSRS for Portview	Software-level source requirements

Source Document	Matrix Role
`PV-SDS-01` SwSDS for Portview	Design allocation source
`PV-STP-01` SwSTP for Portview	Unit procedure source
`PV-TP-01` SwTP for Portview	Integration procedure source
`PV-SV-04` Software Verification Plan	Planned verification context
`PV-SV-05` Software Verification Report	Executed verification context

6.1 Current Direct Links

The following links are currently available within this workspace.

Linked Record	Current Link
System requirements	PV-RS-01
Software requirements	PV-SRS-01
Software design	PV-SDS-01
Unit procedure	PV-STP-01
Unit result	PV-STR-01
Integration procedure	PV-TP-01
Integration result	PV-TR-01
System procedure	PV-SYSTP-01
System result	PV-SYSTR-01
Cybersecurity requirements	PV-CSRS-01
Risks FMEA	FMEA-Z01
Network security enclosure	NSE-Z01
Verification plan	PV-SV-04

Linked Record	Current Link
Verification report	PV-SV-05

7. Primary Consumers

This traceability matrix is used primarily by:

- `PV-SV-01` for lifecycle evidence and release conclusion
- `PV-SV-02` for lifecycle planning and control discussion
- `PV-SV-03` for design allocation context
- `PV-SV-04` for verification planning
- `PV-SV-05` for executed evidence summary